

**Report**  
**Of**  
**Advanced Software Engineering Job Simulation**

Submitted To  
**School of Computer Science and Engineering**  
**IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of  
**B.Tech CSE**



By  
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**Section: 1CSE29**  
**Programme: B.Tech – Computer Science and Engineering**

## Table of Contents

| Chapter                                | Page No. |
|----------------------------------------|----------|
| 1. Student Declaration                 | ii       |
| 2. Acknowledgement                     | iii      |
| 3. Internship / Job Simulation Details | iv       |
| 4. Project Description                 | 1        |
| 4.1 Introduction                       | 1        |
| 4.2 Objectives                         | 1        |
| 4.3 Advanced Data Structures           | 2        |
| 4.4 Software Architecture              | 2        |
| 4.5 Relational Database Design         | 3        |
| 4.6 Data Munging                       | 3        |
| 4.7 Skills and Learning Outcomes       | 4        |
| 4.8 Conclusion                         | 4        |
| 5. References / Supporting Document    | 5        |
| Appendix – Certificate of Completion   | 6        |

## **Student Declaration**

I, Krish Raj, Roll No. 25SCS1003003690, Section 1CSE29, student of IILM UNIVERSITY, GREATER NOIDA, declare that this report presents the work and learning activities associated with the Advanced Software Engineering Job Simulation completed during September 2026. The report has been prepared for academic documentation and reflects the practical task areas stated on the Certificate of Completion.

Student Signature: \_\_\_\_\_

Name: Krish Raj

Roll No.: 25SCS1003003690

Section: 1CSE29

## **Acknowledgement**

I would like to express my sincere gratitude to IILM UNIVERSITY, GREATER NOIDA for providing an academic environment that encourages practical learning and professional development. I am also thankful to Forage for providing the Advanced Software Engineering Job Simulation, which gave me an opportunity to work through practical task areas related to software engineering and computer science. The simulation helped me connect academic concepts with practical problem-solving and gave me a better understanding of the importance of structured thinking in software development.

Krish Raj

Roll No.: 25SCS1003003690

Section: 1CSE29

## **Internship / Job Simulation Details**

|                             |                                                 |
|-----------------------------|-------------------------------------------------|
| <b>Student Name</b>         | Krish Raj                                       |
| <b>Roll No.</b>             | 25SCS1003003690                                 |
| <b>Section</b>              | 1CSE29                                          |
| <b>University</b>           | IILM UNIVERSITY, GREATER NOIDA                  |
| <b>Simulation</b>           | Advanced Software Engineering Job<br>Simulation |
| <b>Period</b>               | September 2026                                  |
| <b>Issuing Organization</b> | Forage                                          |

The certificate states that Krish Raj completed practical tasks during September 2026.

# **Project Description**

## **4.1 Introduction**

The Advanced Software Engineering Job Simulation was completed by Krish Raj during September 2026. According to the Certificate of Completion, the simulation involved practical tasks in four major areas: Advanced Data Structures, Software Architecture, Relational Database Design, and Data Munging.

These areas represent important parts of software engineering. Data structures help in organizing and processing information efficiently. Software architecture focuses on the organization of software components and their interactions. Relational database design deals with structured storage and relationships between data. Data munging involves preparing and transforming data so that it can be used effectively.

The purpose of this report is to document the simulation, summarize the four task areas, and describe the learning outcomes in a simple academic format.

## **4.2 Objectives**

- To gain practical exposure to software-engineering tasks.
- To strengthen problem-solving and analytical thinking.
- To understand the role of advanced data structures in software systems.
- To understand the importance of software architecture.
- To learn the purpose of relational database design.
- To understand the preparation and transformation of data.

### **4.3 Advanced Data Structures**

Advanced data structures are used to organize, store, and process data efficiently. The simulation included Advanced Data Structures as one of its practical task areas, encouraging structured problem solving and consideration of how data is represented and handled.

- Data organization and efficient access.
- Selection of suitable structures for different problems.
- Logical and systematic problem solving.

### **4.4 Software Architecture**

Software architecture describes the high-level structure of a software system. It helps developers understand how major components are arranged and how they communicate. The simulation included Software Architecture as a practical task area.

- Breaking a system into understandable components.
- Defining responsibilities of major components.
- Designing software in a structured and maintainable way.

## **4.5 Relational Database Design**

Relational database design focuses on organizing information into related tables and defining how data items are connected. It is an important part of applications that need reliable and structured data storage. The simulation included Relational Database Design.

- Organizing data into structured tables.
- Identifying relationships between data.
- Reducing unnecessary duplication of information.

## **4.6 Data Munging**

Data munging refers to preparing and transforming data into a useful form. Real-world data may need cleaning, restructuring, or conversion before it can be used effectively. The simulation included Data Munging as a practical task area.

- Preparing data for further use.
- Transforming information into a consistent structure.
- Handling data in a systematic manner.



## **4.7 Skills and Learning Outcomes**

- Improved understanding of practical software-engineering workflows.
- Better structured problem-solving approach.
- Greater awareness of efficient data organization.
- Basic understanding of designing software at a system level.
- Improved understanding of structured relational data.
- Exposure to practical data preparation and transformation.
- Improved confidence in connecting academic concepts with practical tasks.

## **4.8 Conclusion**

The Advanced Software Engineering Job Simulation provided practical exposure to four important areas: Advanced Data Structures, Software Architecture, Relational Database Design, and Data Munging. Completing these task areas helped build a broader understanding of how different parts of software engineering work together.

For Krish Raj, the simulation served as an opportunity to strengthen technical thinking, understand practical software concepts, and document professional learning experience. The Certificate of Completion confirms successful completion of the practical tasks during September 2026.

## 5. References / Supporting Document

1. Certificate of Completion – Advanced Software Engineering Job Simulation, issued by Forage, September 10, 2026.

### Appendix – Certificate of Completion

#### APPENDIX – CERTIFICATE OF COMPLETION





**krish raj**

**Advanced Software Engineering Job Simulation**

Certificate of Completion  
September 10th, 2026

Over the period of September 2026, krish raj has completed practical tasks in:

- Advanced Data Structures
- Software Architecture
- Relational Database Design
- Data Munging

  
**Suresh Kumar**  
Global Chief  
Technology Officer and  
Chief Development  
Officer

  
**Tom Brunskill**  
Co-Founder of Forage

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